Signals and Systems
SET214
Lecture (03)- Introductory Concepts

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Outlines

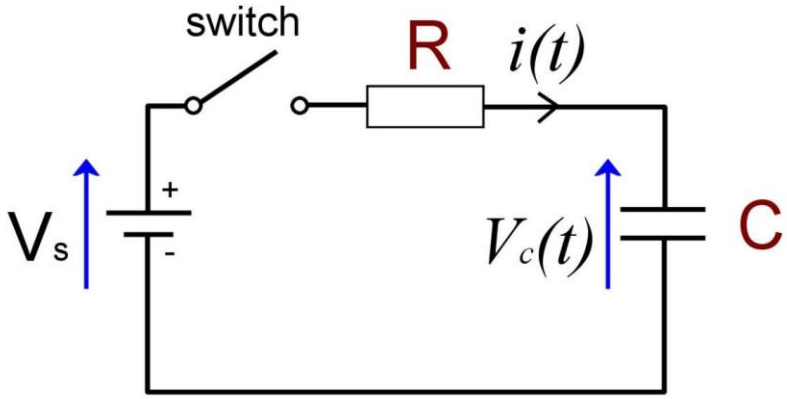
- Signals
- Classification of Signals
- CT-Elementary Functions
- Properties of Delta Function

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- Signals
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- Properties of Delta Function

Signals (1/2)

- A signal is a function of one or more variables that conveys information about some (usually physical) phenomenon.
- For a function *f* , in the expression $f(t_1,t_2,\ldots,t_n)$,
  - *t* is called an independent variable
  - while the function value itself, *f* ,is referred to as a dependent variable.
- Some examples of signals include
  - voltage or current in an electronic circuit.
  - position, velocity, or acceleration of an object.
  - force or torque in a mechanical system.
  - flow rate of a liquid or gas in a chemical process.
  - digital image, digital video, or digital audio.

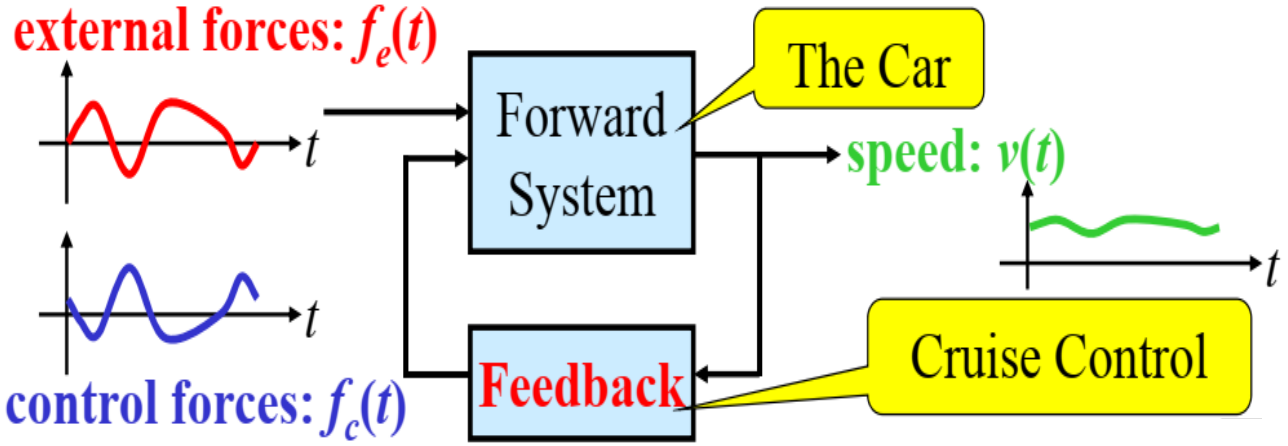


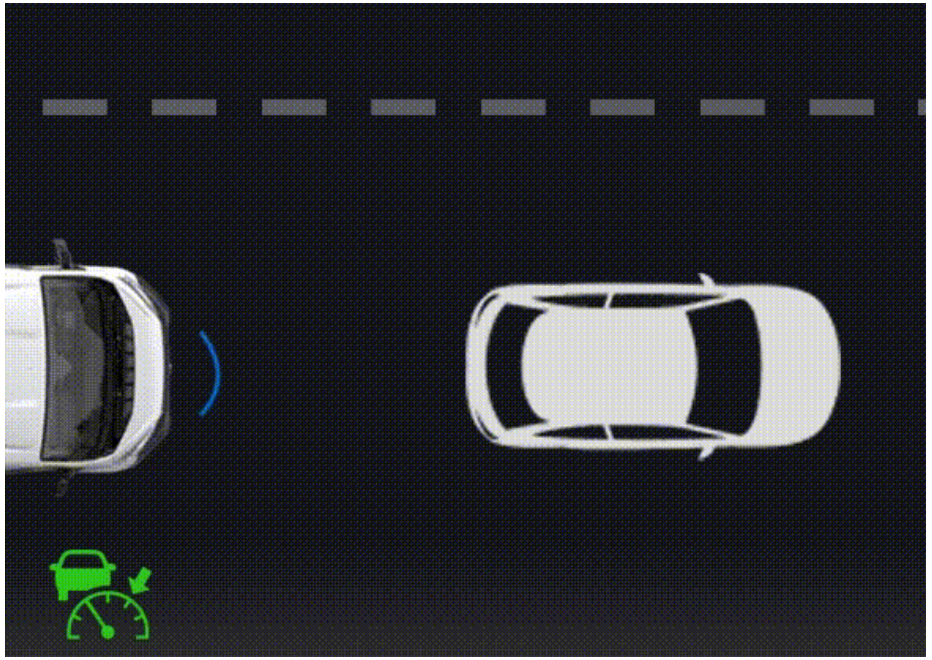
$V_R(t) = i(t)*R$

$V_c(t) = \frac{1}{C}\int i(t)dt$

Signals (2/2)

- Design a system that gives a desired output
  Ex., Control System…. such as cruise control for a car





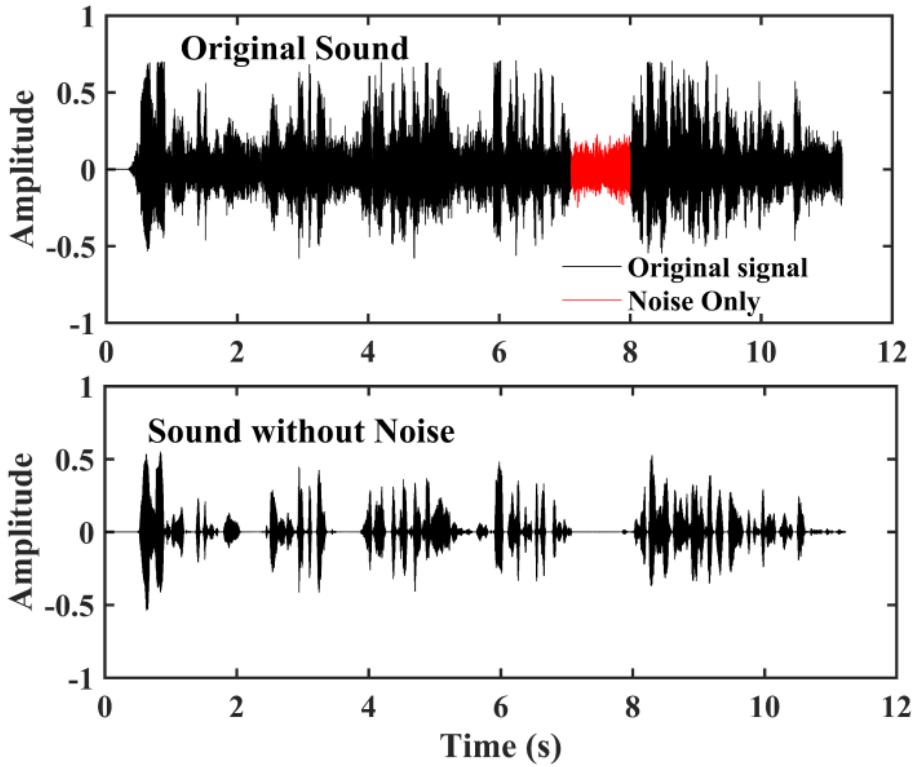
Adaptive cruise control

Outlines

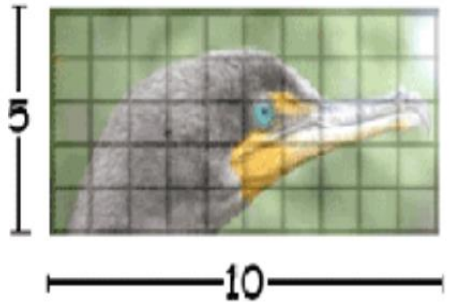
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Classification of Signals (1/3)

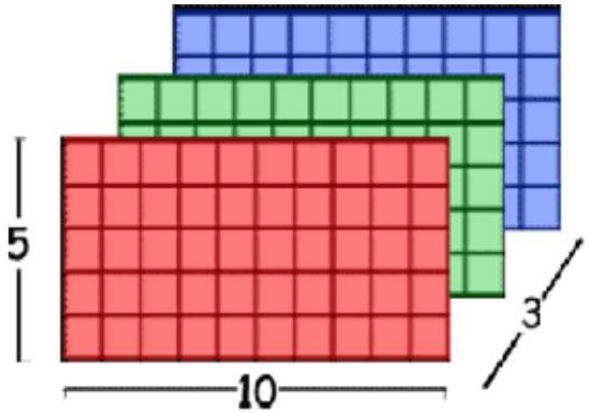
- Number of independent variables (i.e., dimensionality):
  - A signal with *one* independent variable is said to be one dimensional (e.g., audio Signal).
  - A signal with *more than one* independent variable is said to be multi-dimensional (e.g., image).



Audio Signal



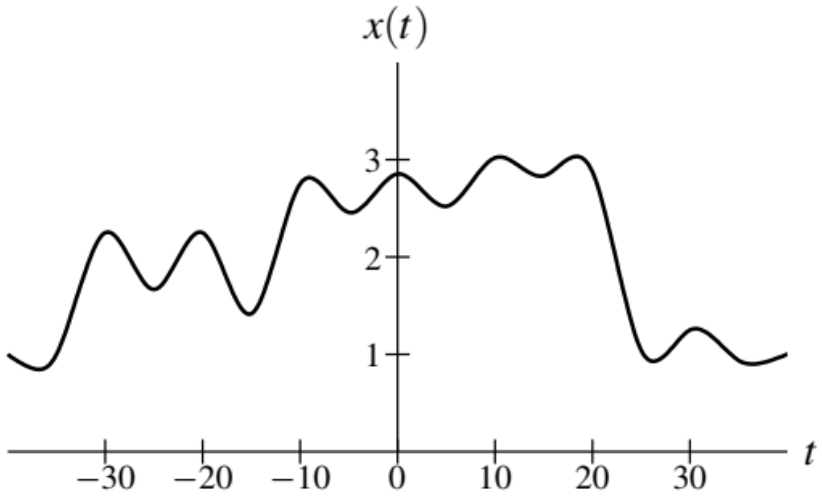
Original Color Image



RGB Matrix

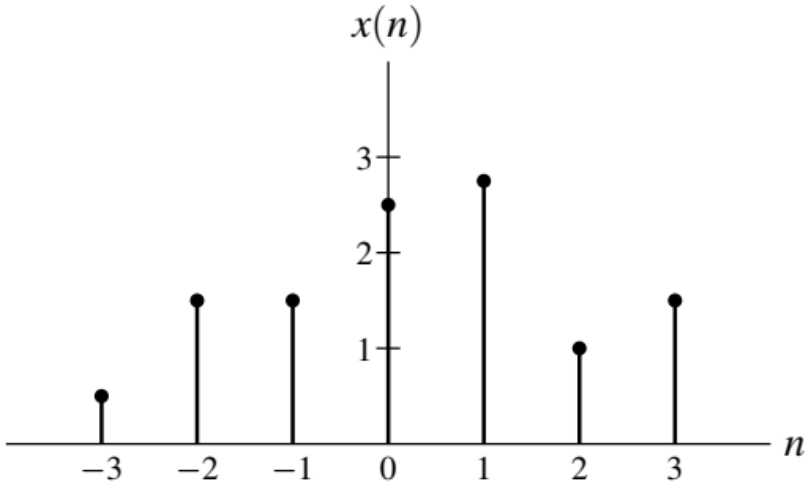
Classification of Signals (2/3)

- **Continuous or discrete variables:**
  - A signal with *continuous* independent variables is said to be Continuous-Time (CT) (e.g., voltage waveform)
  - A signal with *discrete* independent variables is said to be Discrete-Time (DT) (e.g., digital image, tock market index).



Continuous-Time (CT) Signal

CT signal is said to be analog (e.g., voltage waveform)

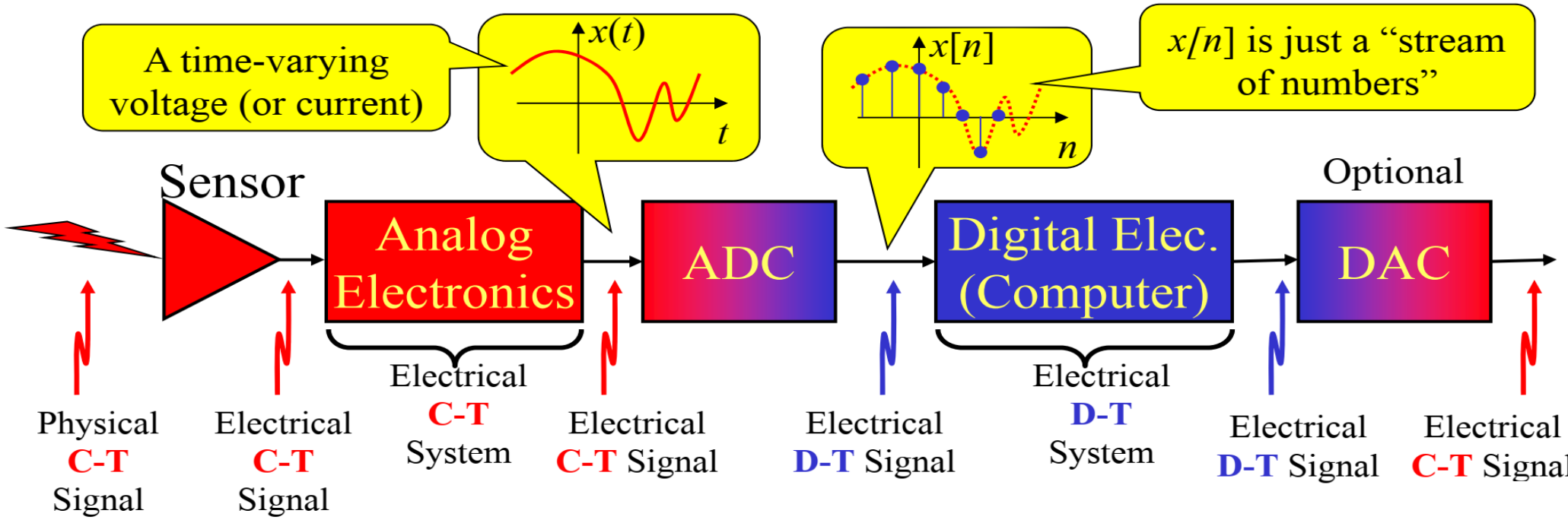


Discrete-Time (DT) Signal

DT signal is said to be digital (e.g., digital audio/image)

Classification of Signals (3/3)

- **Continuous or discrete variables:**
  - A signal with *continuous* independent variables is said to be Continuous-Time (CT) (e.g., voltage waveform)
  - A signal with *discrete* independent variables is said to be Discrete-Time (DT) (e.g., digital image, tock market index).

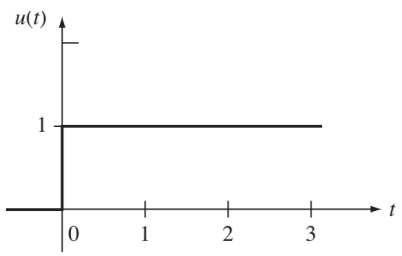


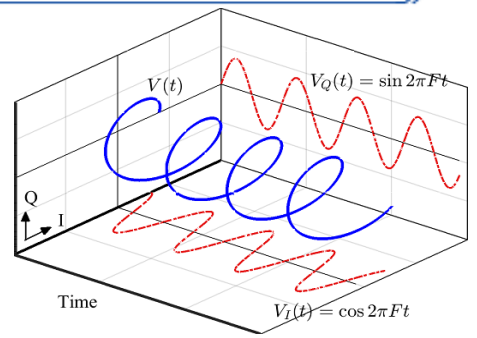
Outlines

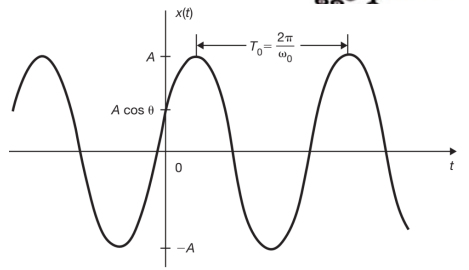
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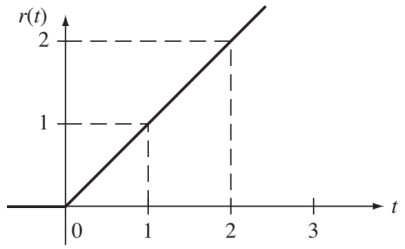
CT-Elementary Functions

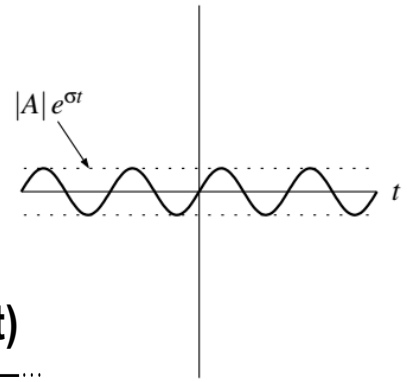
- **Continuous** Elementary Functions
  - Real Sinusoidal Function
  - Real Exponential Function
  - Complex Sinusoidal Function
  - Complex Exponential Function
  - Unit-Step Function
  - Ramp Function
  - Signum Function
  - rectangle Function
  - Triangular Function
  - Sinc Function
  - Unit-Impulse Function

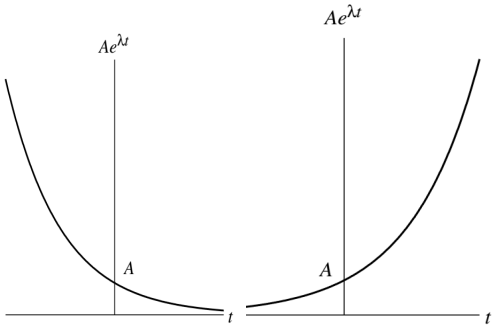


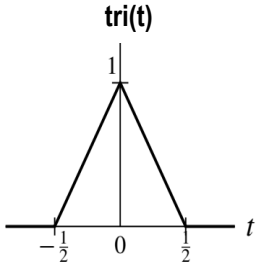


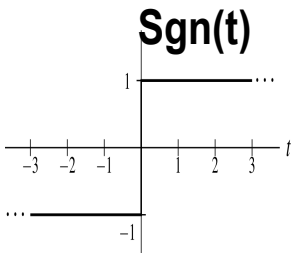


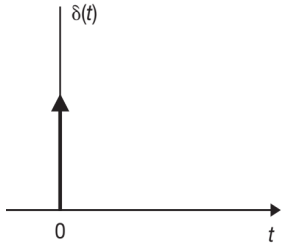


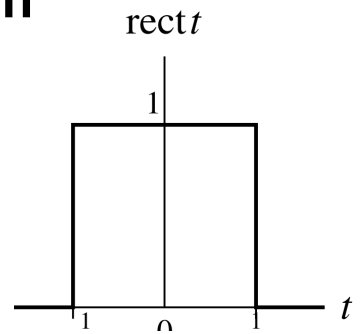


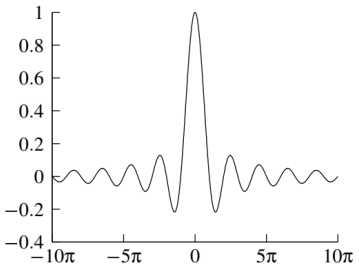


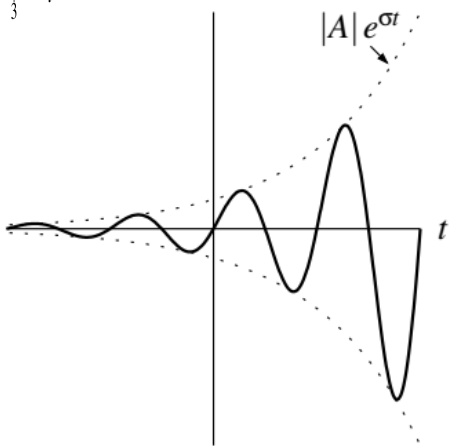


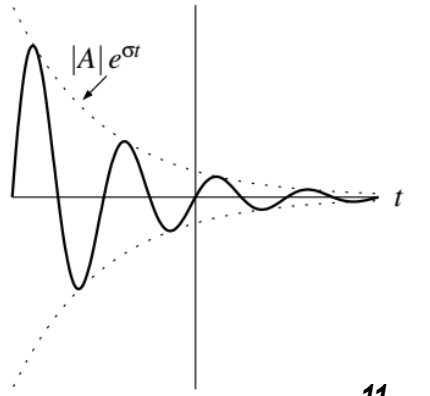












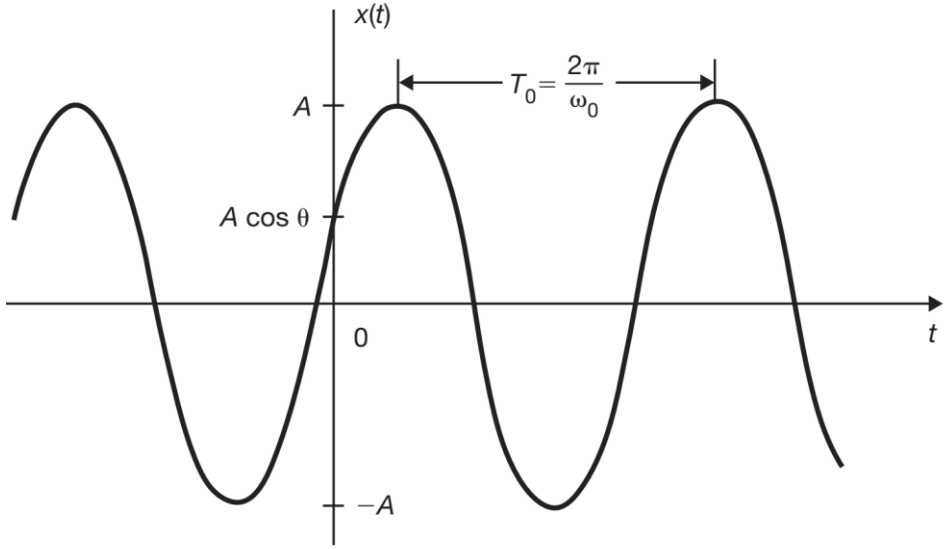
Real Sinusoidal Function

- A real sinusoidal function is a function of the form:- $x(t) = A\cos(\omega_0 t + \theta)$

A: is the amplitude (real)

ω: is the radian frequency in radians per second

θ: is the phase angle in radians



Continuous-time sinusoidal signal

- X(t) is periodic with fundamental period

$$T_0 = \frac{2\pi}{\omega_0}$$

- The reciprocal of the fundamental period T_0 is called the fundamental frequency f_0:

$$f_0 = \frac{1}{T_0} \quad \text{hertz (Hz)}$$

- Fundamental angular frequency

$$\omega_0 = 2\pi f_0$$

Real Exponential Function

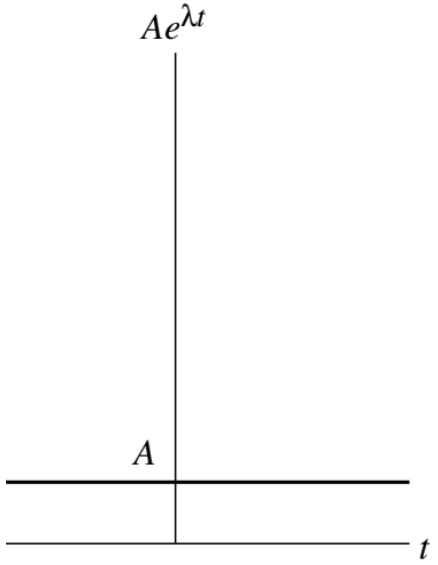
- **Exponential** function is a function of the form:-

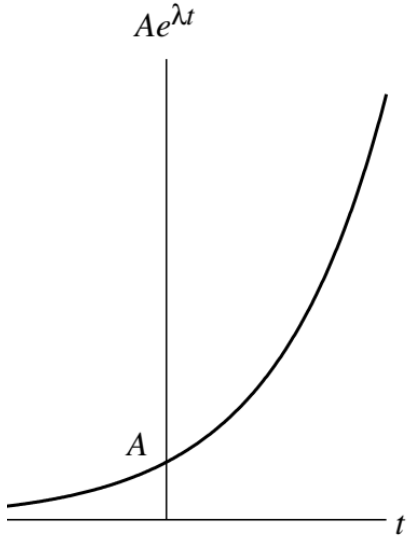
$$x(t) = Ae^{\lambda t}$$

- A **real exponential** function when A and λ are restricted to be *real* numbers.
- A **real exponential** can exhibit one of *three distinct modes* of behavior, depending on the value of λ.

1

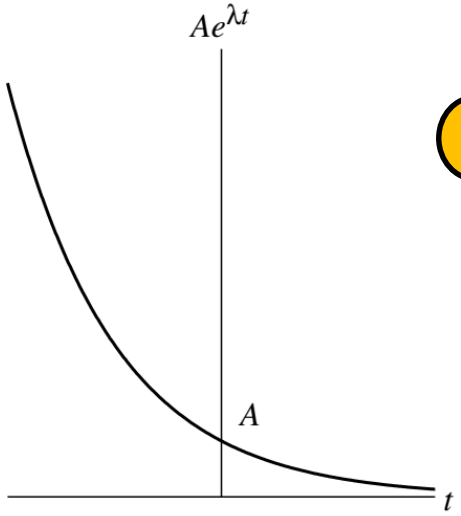
If $\lambda = 0$, $x(t)$ simply equals the *constant* A





2

If $\lambda > 0$, $x(t)$ *increases* exponentially as t increases (i.e., a growing exponential).



3

If $\lambda < 0$, $x(t)$ *decreases* exponentially as t increases (i.e., a decaying exponential).

Complex Sinusoidal Function

- **Exponential** function is a function of the form:-

$$x(t) = Ae^{\lambda t}$$

- A **complex sinusoidal** function is a **special case of a complex exponential** $x(t) = Ae^{\lambda t}$, where A is **complex** and λ is **purely imaginary** (i.e., $Re\{\lambda\} = 0$).
- That is, a complex sinusoidal function is a function of the form:-

$$x(t) = Ae^{j\omega t}$$

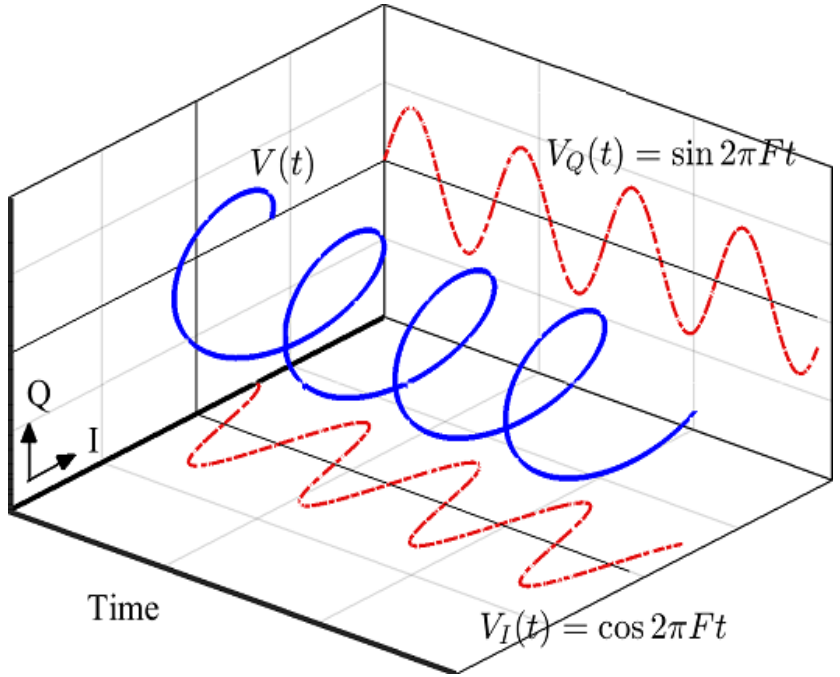
A is *complex*

- By expressing A in polar form as $A = |A|e^{j\theta}$ (where θ is real)

$$x(t) = |A|\ e^{j\theta} * e^{jwt} \quad \Rightarrow \quad x(t) = |A|\ e^{j(wt+\theta)}$$

- Using **Euler's** relation, we can rewrite $x(t)$ as

$$x(t) = \underbrace{|A|\cos(\omega t + \theta)}_{Re\{x(t)\}} + j\underbrace{|A|\sin(\omega t + \theta)}_{Im\{x(t)\}}$$



Remember

From **Euler's relation**, a complex sinusoid can be expressed as the sum of two real sinusoids as

$$Ae^{j\omega t} = A\cos(\omega t) + jA\sin(\omega t)$$

Complex Exponential Function (1/2)

- Exponential function is a function of the form:-

$$x(t) = Ae^{\lambda t}$$

- In the most general case of a complex exponential function $x(t) = Ae^{\lambda t}$, A and λ are both *complex :-*

$A = |A|e^{j\theta}$ and $\lambda = \sigma + j\omega$

$$x(t) = |A|\ e^{j\theta} * e^{(\sigma + j\omega)t}$$

$$= |A|\ e^{j\theta} * e^{\sigma t}\ e^{j\omega t}$$

$$= |A|\ e^{\sigma t} * e^{j(\omega t + \theta)}$$

- Using Euler's relation, we can rewrite $x(t)$ as

$$x(t) = \underbrace{|A|\, e^{\sigma t}\cos(\omega t + \theta)}_{\mathrm{Re}\{x(t)\}} + j\underbrace{|A|\, e^{\sigma t}\sin(\omega t + \theta)}_{\mathrm{Im}\{x(t)\}}$$

General Complex Exponential Functions

One of *three distinct modes* of behavior is exhibited by $x(t)$, depending on the value of σ

Remember

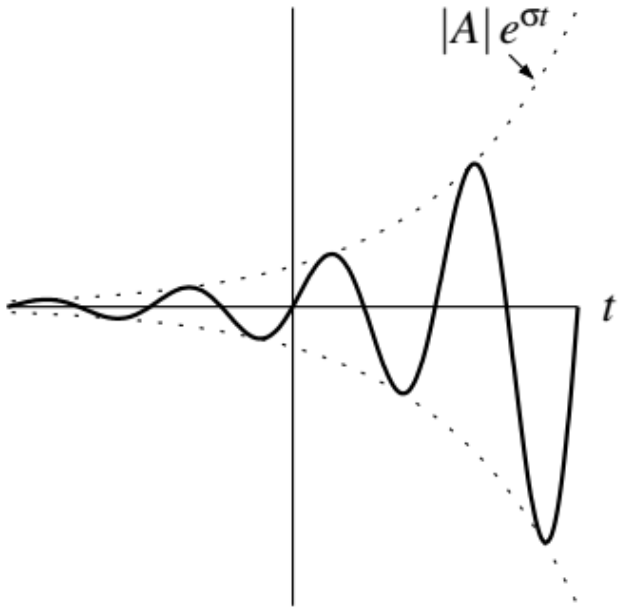
From Euler's relation, a complex sinusoid can be expressed as the sum of two real sinusoids as

$$Ae^{j\omega t} = A\cos(\omega t) + jA\sin(\omega t)$$

Complex Exponential Function (2/2)

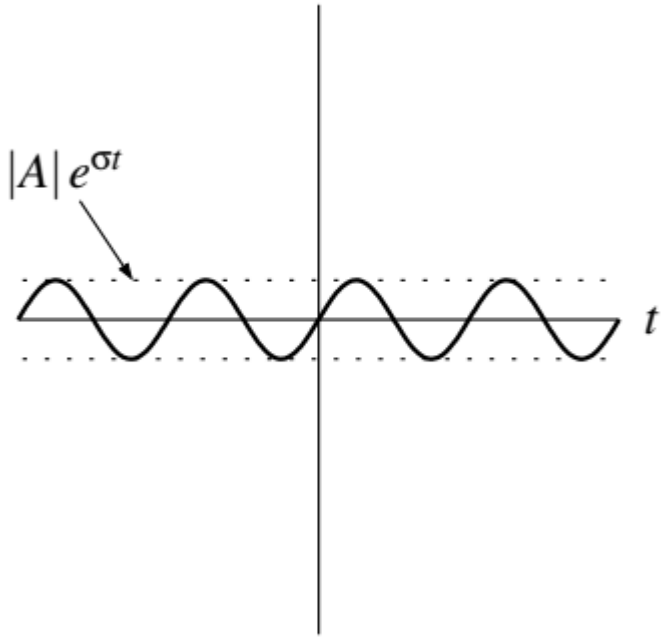
❑ The three modes of behavior for $x(t) = \underbrace{|A|e^{\sigma t}\cos(\omega t+\theta)}_{\mathrm{Re}\{x(t)\}} + j\underbrace{|A|e^{\sigma t}\sin(\omega t+\theta)}_{\mathrm{Im}\{x(t)\}}$

Growing sinusoidal signal



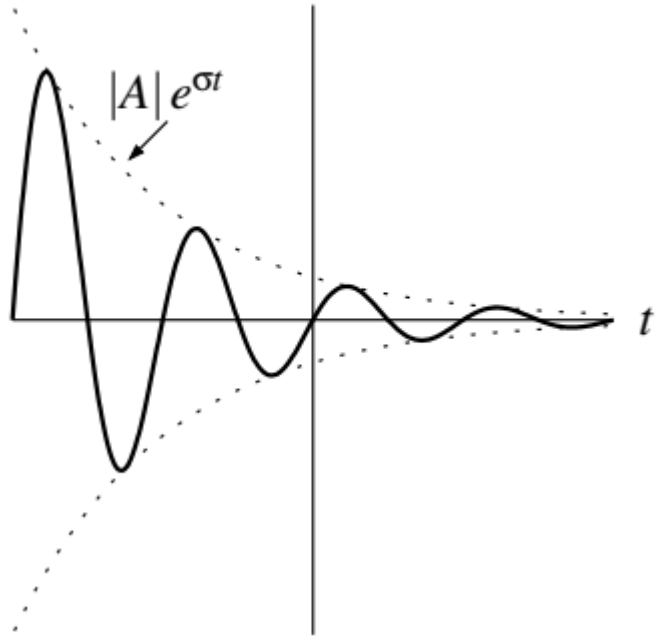
If $\sigma > 0$, Re{x} and Im{x} are each the product of a real sinusoid and a growing real exponential.

Sinusoidal signal



If $\sigma = 0$, Re{x} and Im {x} are real sinusoids

Decaying sinusoidal signal

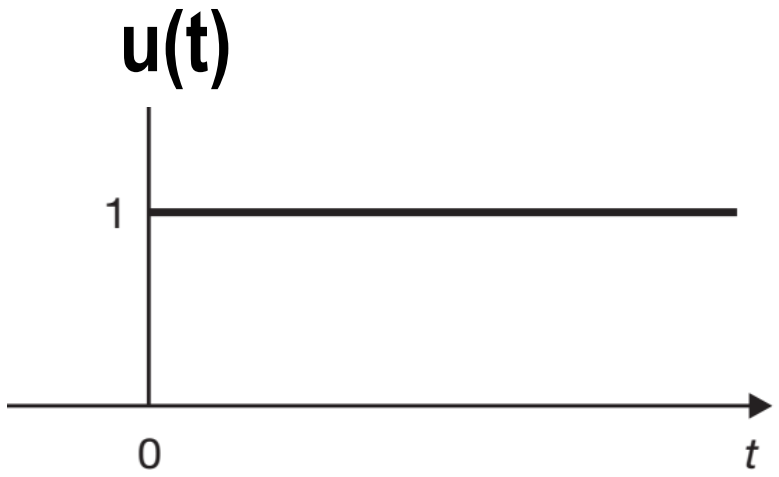


If $\sigma < 0$, Re{x} and Im{x} are each the product of a real sinusoid and a decaying real exponential.

Unit-Step Function

- The unit-step function (also known as the Heaviside function), denoted *u*, is defined as:-

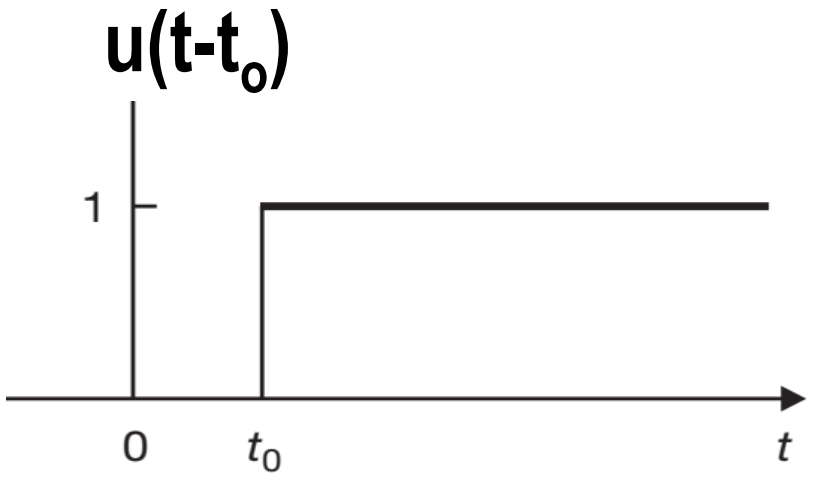
$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



- Here unit step means that the amplitude of u(t) is equal to 1 for all t≥0

- Note , the shifted unit step function $u(t-t0)$ is defined as:-

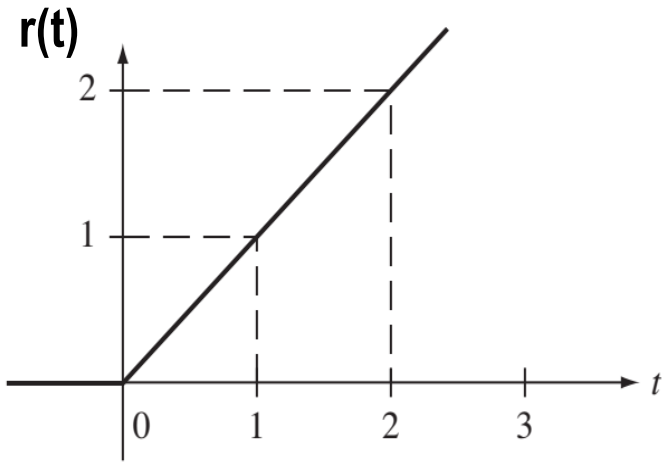
$$u(t-t_0) = \begin{cases} 1 & t \geq t_0 \\ 0 & t < t_0 \end{cases}$$



Ramp Function

- The unit-ramp function r(t) is defined mathematically by :-

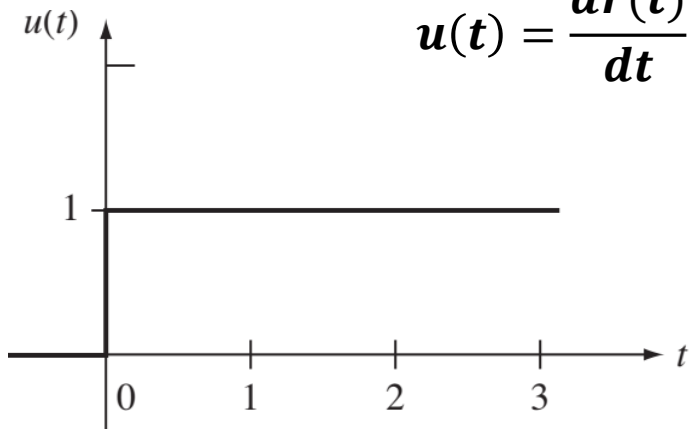
$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



- Note that for the slope of $r(t)$ is 1.
- Thus $r(t)$ has "unit slope," which is the reason $r(t)$ is called the unit-ramp function

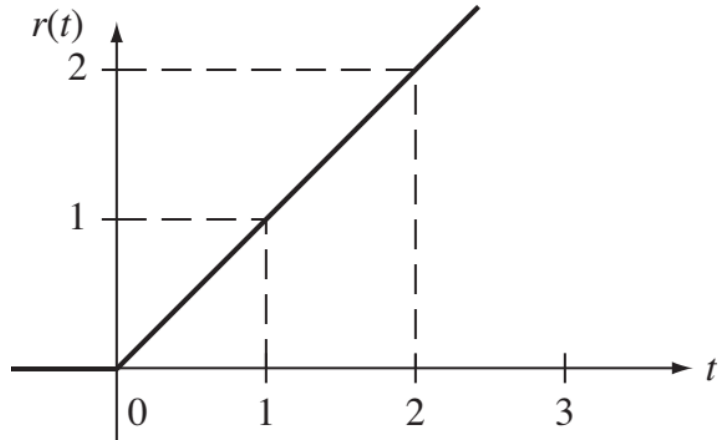
- Relation between step and ramp functions

$$u(t) = \frac{dr(t)}{dt} = 1$$



Integration

Differentiation

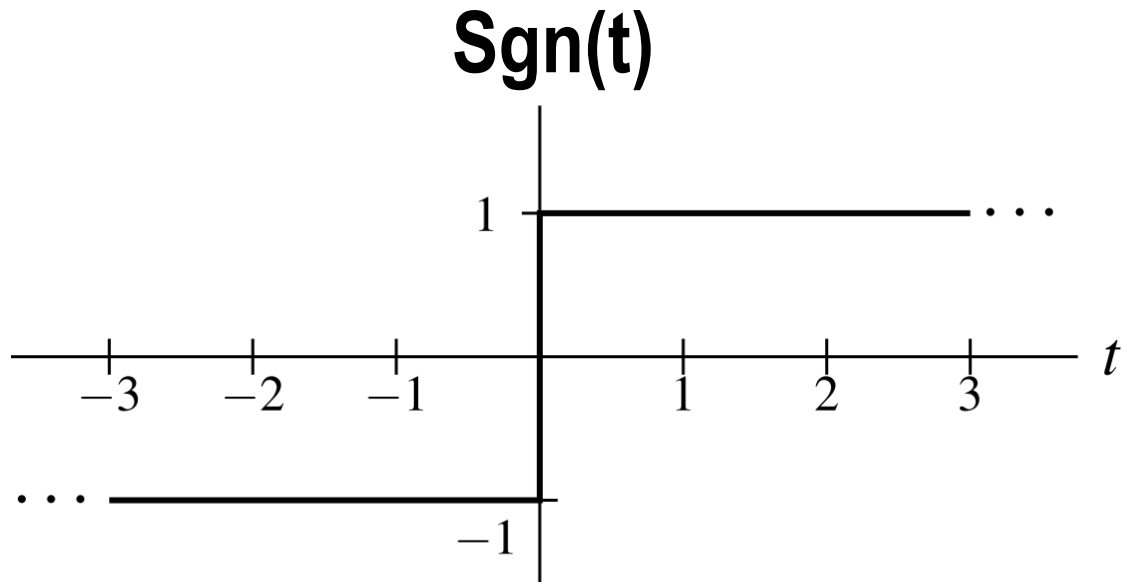


$$r(t) = \int u(t)\ dt = \mathrm{t}$$

Signum Function

- The signum function, denoted sgn, is defined as

$$\text{Sgn}(t) = \begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases}$$

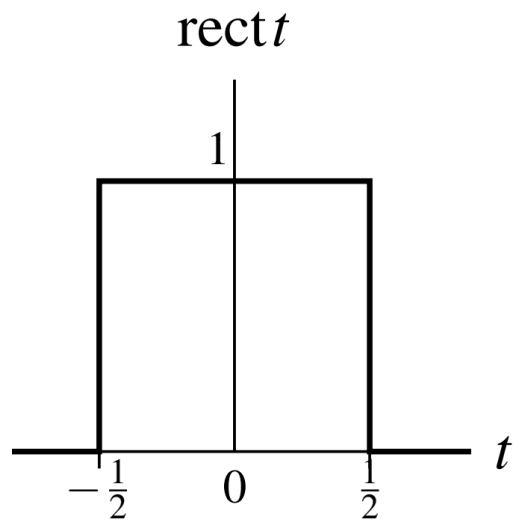


From its definition, one can see that the signum function simply computes the *sign* of a number.

Rectangular Function

- The rectangular function (also called the unit-rectangular pulse function), denoted *rect*, is given by:-

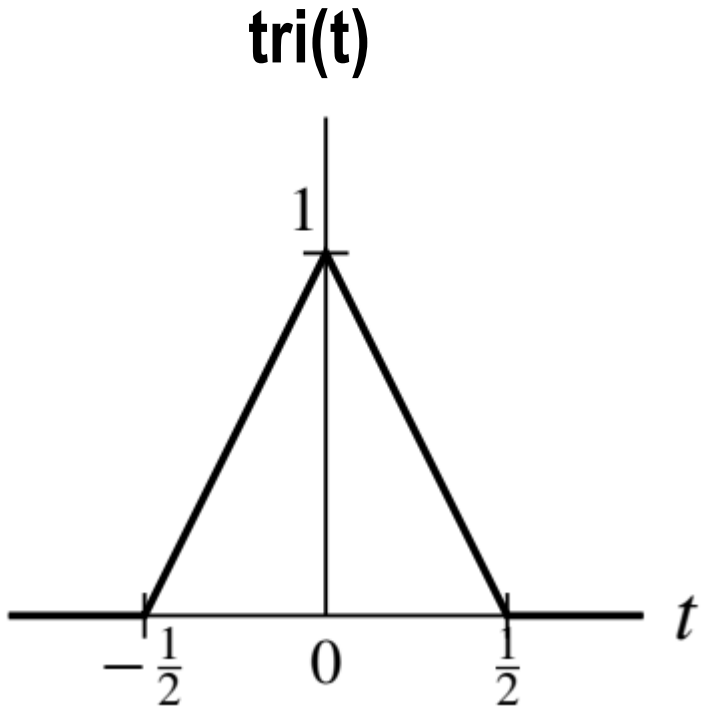
$$\text{rect(t)} = \begin{cases} 1 & -\frac{1}{2} \le t < \frac{1}{2} \\ 0 & \text{otherwise.} \end{cases}$$



Triangular Function

- The triangular function (also called the unit-triangular pulse function), denoted tri (t), is defined as:-

$$\text{tri}(t) = \begin{cases} 1 - 2|t| & |t| \le \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

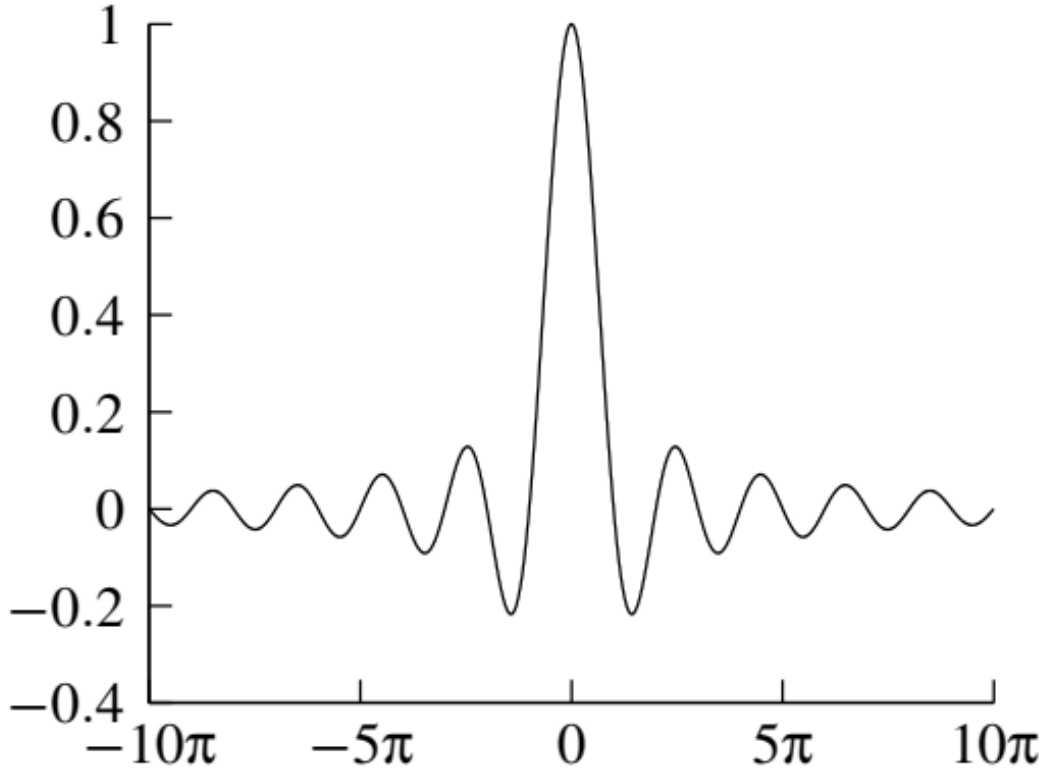


The unit triangle is related to the unit rectangle through an operation called convolution

Sinc Function

- The unit Sinc function is related to the unit Rectangle function through the Fourier Transform.
- It is used for noise removal in signals

$$\operatorname{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$



Noted:-→ By l'Hopital's rule,

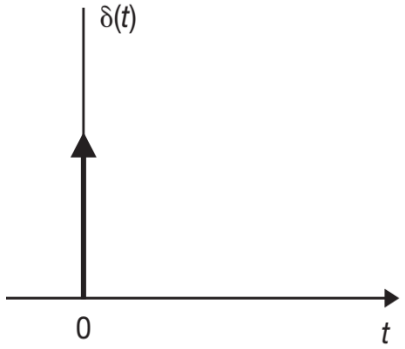
sinc(0) = 1

Unit-Impulse Function

- The unit impulse $\delta(t)$ also called the delta function or the Dirac distribution, is defined by:-

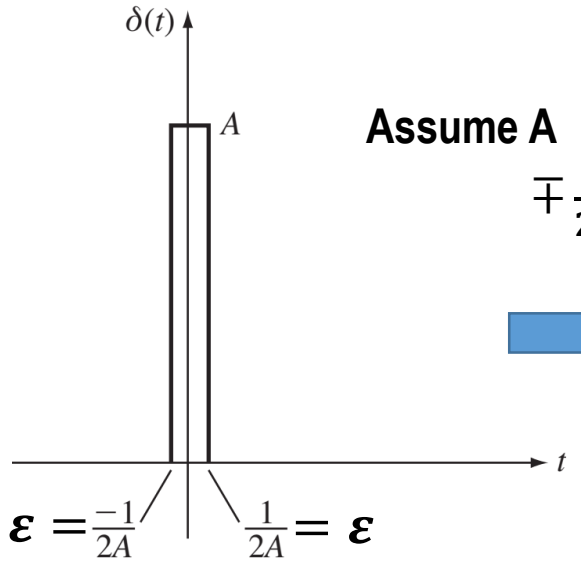
$$\delta(t) = \begin{cases} 0 & t \neq 0 \\ \infty & t = 0 \end{cases}$$

$$\int_{-\varepsilon}^{\varepsilon} \delta(t)\, dt = 1$$



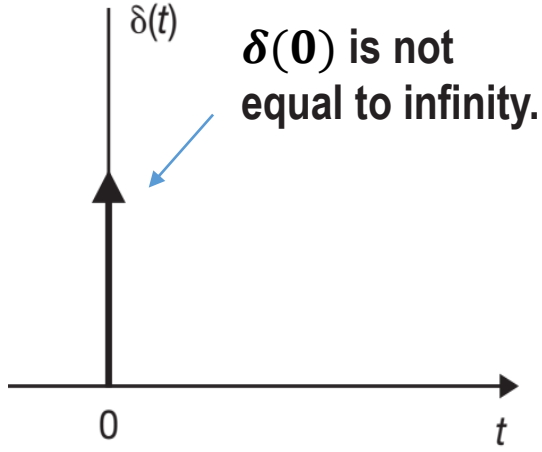
- The 1st condition states that $\delta(t)$ is zero for all nonzero values of *t*, while the 2nd condition states that the area under the impulse is 1, so $\delta(t)$ has unit area.

- The impulse $\delta(t)$ can be approximated by a pulse centered at the origin with amplitude *A* and time duration 1/*A*, where *A* is a very large positive number .



Assume A → ∞

∓ 1/2A → 0



$\delta(0)$ is not equal to infinity.

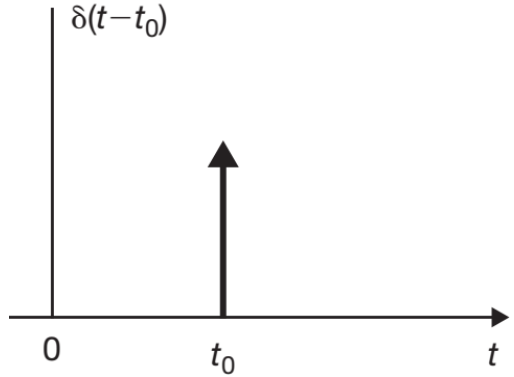
Unit impulse function

$$\int_{-\infty}^{\infty} \delta(t)\, dt = \int_{\frac{-1}{2A}}^{\frac{1}{2A}} A\, dt$$

$$= A * t \Big|_{\frac{-1}{2A}}^{\frac{1}{2A}}$$

$$= A\left(\frac{1}{2A} - \left(\frac{-1}{2A}\right)\right)$$

$$= A\left(\frac{1}{A}\right) = 1$$



shifted unit impulse function

Outlines

- Signals
- Classification of Signals
- CT-Elementary Functions
- Properties of Delta Function

Properties of Delta Function (1/3)

❑ Equivalence property:

$$x(t)\delta(t) = x(0)\delta(t)$$

$$x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$$

❑ Scaling property:

$$\delta(at) = \frac{1}{|a|}\delta(t)$$

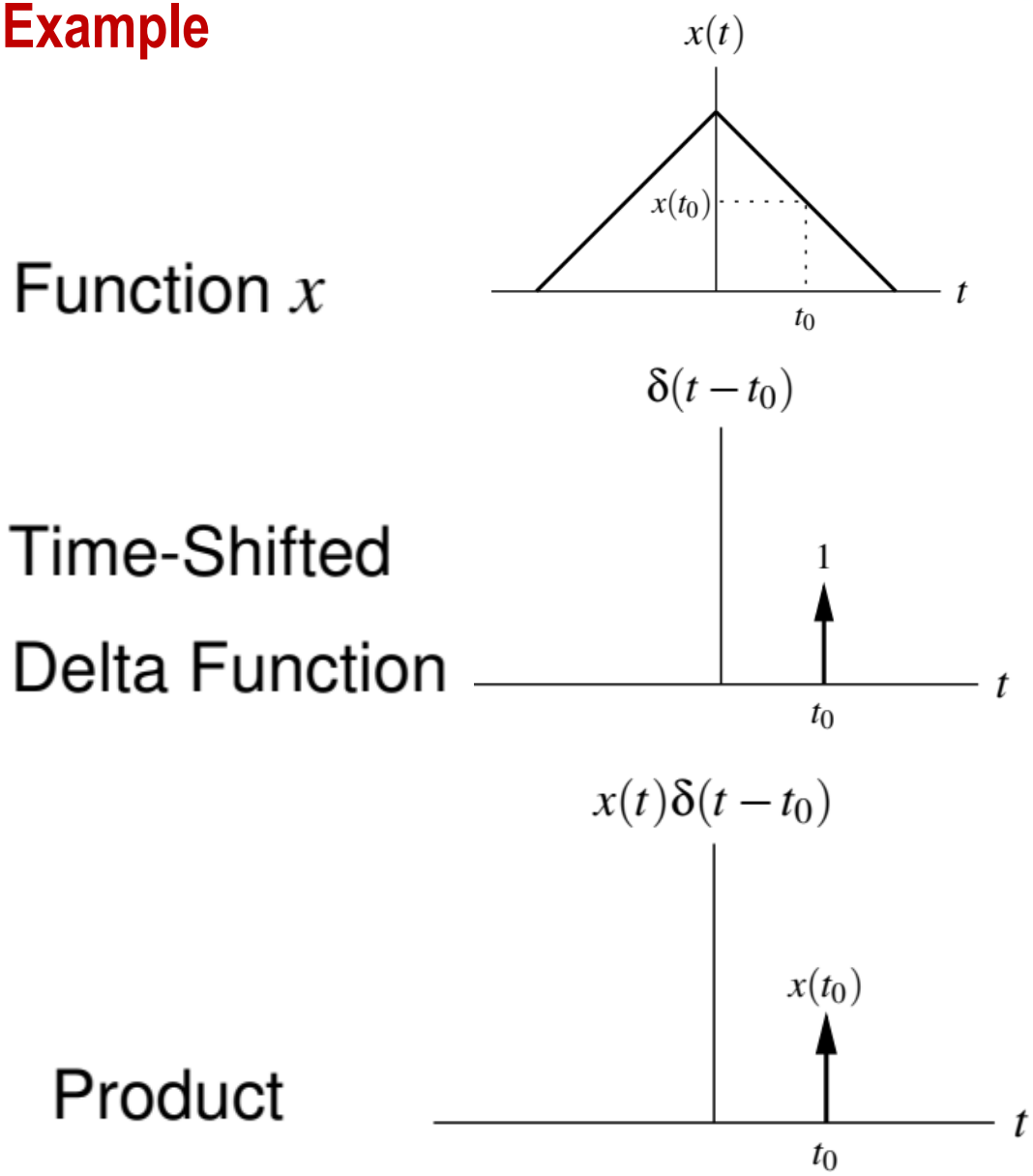
❑ Even property:

$$\delta(t) = \delta(-t)$$

❑ Sifting property:

$$\int_{-\infty}^{\infty} x(t)\delta(t-t_0)dt = x(t_0)$$

❑ Example



Properties of Delta Function (2/3)

- Q1) If x(t)= (2t+5) find f(t) =x(t) $\delta(t)$

Ans.

f(t) =x(t) $\delta(t)$ = (2t+5) @ t=0

= 2*0+5=5

- Q2) If x(t)= (2t+5) find f(t)=x(t) $\delta(t-1)$

Ans.

f(t)=x(t) $\delta(t-1)$ = (2t+5) @ t=1

= 2*1+5=7

- Q3) If x(t)= 2sin(t) find f(t) =x(t) $\delta(t-\frac{\pi}{2})$

Ans.

f(t) =x(t) $\delta\left(t-\frac{\pi}{2}\right)$ =2sin(t) $\delta\left(t-\frac{\pi}{2}\right)$ @ t= $\frac{\pi}{2}$

=2 sin($\frac{\pi}{2}$)=2

- Q4) find $\int_{-\infty}^{\infty} \sin(t)\delta\left(t-\frac{\pi}{4}\right)dt$

Ans.

Solution. Using the sifting property of the delta function, we have

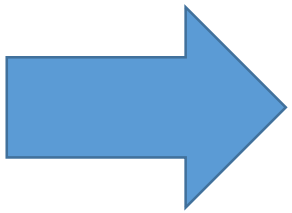
$$\int_{-\infty}^{\infty} \sin(t)\delta\left(t-\frac{\pi}{4}\right)dt = \sin\left(\frac{\pi}{4}\right)$$

$$= \frac{1}{\sqrt{2}}.$$

Properties of Delta Function (2/3)

Evaluate the integral

$$\int_{-\infty}^{\infty} \sin(2\pi t)\delta(4t-1)dt$$



Try to solve it in Section

Thanks!